

# Preface

Thank you very much for choosing our company's products. We are dedicated to providing you with the best and highest quality service.

This manual may contain some errors or inaccuracies, and we apologize for any inconvenience this may cause. We appreciate your valuable feedback, and we will regularly update the manual's content.

If you have any questions about the use of this manual and SDK, please contact our technical support team. To ensure a quick resolution of issues, please provide the following information before communication:

1. The current system platform using this SDK, such as Windows, Linux, or others, and specify the platform's detailed version number and bit (32-bit or 64-bit).
2. The model of our hardware product currently in use.
3. The version number of the SDK currently in use.
4. Detailed description of the issue, including, but not limited to, involved functional interfaces, specific error messages, etc.

Thank you for your support!

## Revision History

| Version No. | Release Date | Descriptions    |
|-------------|--------------|-----------------|
| V2.0.0      | 2025-1-21    | Initial release |

# **Chapter 1    Brief Introduction**

Welcome to the SDK\_USB Development Manual. This document provides a detailed description of the various functional interfaces, structures, constants, and the calling process of the SDK interface functions included in our company's USB port infrared device development kit. The main features of the SDK include: device connection, live view, retrieval of temperature data, retrieval of exception data, device configuration, and other functions.

# Chapter 2 Development Guide

## 1. Supported Systems for the SDK

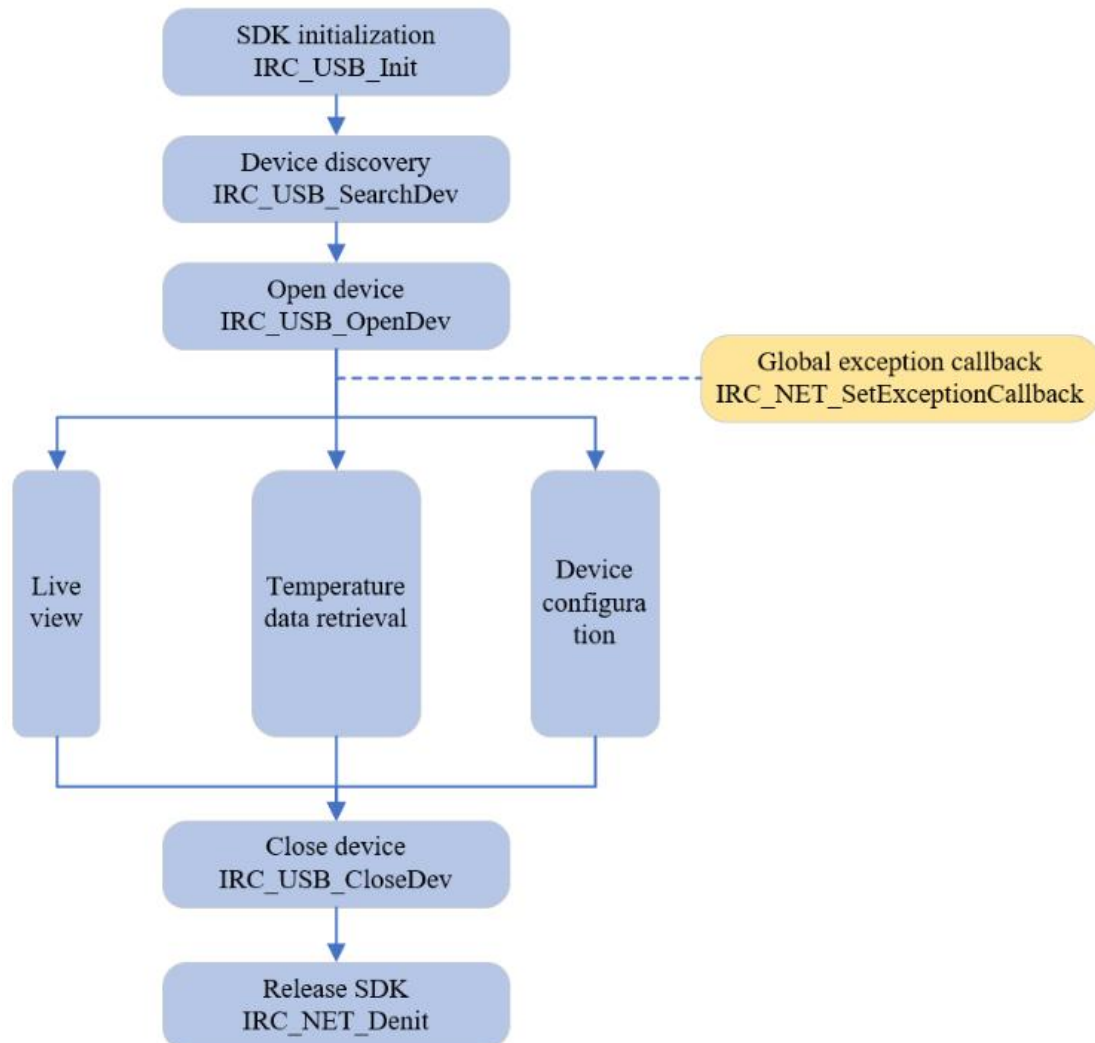
|                                           |
|-------------------------------------------|
| <b>Windows 32/64bit Version:</b>          |
| Windows 11/Windows 10/Windows 8/Windows 7 |
| <b>Linux 32/64bit Version:</b>            |
| Ubuntu 18.04 and above                    |

## 2. Device Compatibility Information

This SDK is compatible with the following types of devices from our company:

| Device Models   |
|-----------------|
| Turing L Series |
| Turing A Series |
| Turing M Series |

### 3. Programming Process



## Chapter 3 Interface List

### 1. Device Connection

| Interface Name                              | Interface Description    |
|---------------------------------------------|--------------------------|
| <a href="#">IRC_USB_Init</a>                | SDK Initialization       |
| <a href="#">IRC_USB_Deinit</a>              | Release SDK              |
| <a href="#">IRC_USB_SearchDev</a>           | Device Discovery         |
| <a href="#">IRC_USB_DEV_SEARCH_CALLBACK</a> | Search Callback Function |
| <a href="#">IRC_USB_OpenDev</a>             | Open Device              |
| <a href="#">IRC_USB_CloseDev</a>            | Close Device             |

#### 1) SDK Initialization IRC\_USB\_Init

| Options      | Introduction                                                  |
|--------------|---------------------------------------------------------------|
| Description  | SDK Initialization                                            |
| Function     | <code>int IRC_USB_Init(<br/>);</code>                         |
| parameter    |                                                               |
| Return Value | (Refer to the <a href="#">status code</a> table for details.) |
| Note         |                                                               |

## 2) Release SDK IRC\_USB\_Deinit

| Options      | Introduction                      |
|--------------|-----------------------------------|
| Description  | Release SDK                       |
| Function     | <pre>int IRC_USB_Deinit( );</pre> |
| parameter    |                                   |
| Return Value |                                   |
| Note         |                                   |

## 3) Device Discovery IRC\_USB\_SearchDev

| Options      | Introduction                                                                                                   |
|--------------|----------------------------------------------------------------------------------------------------------------|
| Description  | Device discovery                                                                                               |
| Function     | <pre>Int IRC_USB_SearchDev( <a href="#">IRC_USB_DEV_SEARCH_CALLBACK</a> searchCallback, void* userData);</pre> |
| parameter    | <p>param[in] searchCallback Search callback function</p> <p>param[in] userData User-defined data</p>           |
| Return Value | <a href="#">status code</a>                                                                                    |
| Note         | For specific usage methods, please refer to the demo                                                           |

#### 4) Search Callback Function

##### IRC\_USB\_DEV\_SEARCH\_CALLBACK

| Options      | Introduction                                                                                       |
|--------------|----------------------------------------------------------------------------------------------------|
| Description  | Search callback function                                                                           |
| Function     | IRC_USB_DEV_SEARCH_CALLBACK(<br><a href="#">IRC_USB_DEV_INFO</a> * searchInfo,<br>void* userData); |
| parameter    | param[in] searchCallback Searched device<br>information<br>param[in] userData User-defined data    |
| Return Value | <a href="#">status code</a>                                                                        |
| Note         | Use in conjunction with the "Device Search"<br>function.                                           |

#### 5) Open Device IRC\_USB\_OpenDev

| Options     | Introduction                                                |
|-------------|-------------------------------------------------------------|
| Description | Open device                                                 |
| Function    | int IRC_USB_OpenDev(<br>int id,<br>IRC_USB_HANDLE* handle); |
| parameter   | param[in] id Device ID, refer to                            |

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
|              | <a href="#">IRC_USB_DEV_INFO</a><br>param[out] handle Operation handle |
| Return Value | <a href="#">status code</a>                                            |
| Note         |                                                                        |

## 6) Close Device    IRC\_USB\_CloseDev

| Options      | Introduction                                     |
|--------------|--------------------------------------------------|
| Description  | Close device                                     |
| Function     | int IRC_USB_CloseDev(<br>IRC_USB_HANDLE handle); |
| parameter    | param[in] handle Operation handle                |
| Return Value | <a href="#">status code</a>                      |
| Note         |                                                  |

## 1. Live View

| Interface Name                         | Interface Description   |
|----------------------------------------|-------------------------|
| <a href="#">IRC_USB_StartPreview</a>   | Start Live View         |
| <a href="#">IRC_USB_VIDEO_CALLBACK</a> | Video Callback Function |
| <a href="#">IRC_USB_StopPreview</a>    | Stop Live View          |

### 1) Start Live View    IRC\_USB\_StartPreview

| Options | Introduction |
|---------|--------------|
|---------|--------------|



|              |                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Description  | Start live view                                                                                                                            |
| Function     | <pre>int IRC_USB_StartPreview(     IRC_USB_HANDLE handle,     <a href="#">IRC_USB_VIDEO_CALLBACK</a> videoCb,     void* userData);</pre>   |
| Parameter    | <p>param[in] handle Operation handle</p> <p>param[in] videoCb video callback function</p> <p>param[in] userData user self-defined data</p> |
| Return Value | <a href="#">status code</a>                                                                                                                |
| Note         | For specific usage methods, please refer to the demo                                                                                       |

## 2) Video Callback Function [IRC\\_USB\\_VIDEO\\_CALLBACK](#)

|              |                                                                                                                                              |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Options      | Introduction                                                                                                                                 |
| Description  | video callback function                                                                                                                      |
| Function     | <pre>Int IRC_USB_VIDEO_CALLBACK(     IRC_USB_HANDLE handle,     <a href="#">IRC_USB_VIDEO_INFO_CB</a>* videoInfo,     void* userData);</pre> |
| Parameter    | <p>param[in] handle Operation handle</p> <p>param[in] videoInfo video callback info</p> <p>param[in] userData user self-defined data</p>     |
| Return Value | <a href="#">status code</a>                                                                                                                  |

|      |                                                         |
|------|---------------------------------------------------------|
| Note | Use in conjunction with the "Start live view" function. |
|------|---------------------------------------------------------|

### 3) Stop Live View IRC\_USB\_StopPreview

| Options      | Introduction                                        |
|--------------|-----------------------------------------------------|
| Description  | Stop live view                                      |
| Function     | int IRC_USB_StopPreview(<br>IRC_USB_HANDLE handle); |
| Parameter    | param[in] handle Operation handle                   |
| Return Value | <a href="#">status code</a>                         |
| Note         |                                                     |

## 2. Temperature Data Retrieval

| Interface Name                        | Interface Description                      |
|---------------------------------------|--------------------------------------------|
| <a href="#">IRC_USB_StartPullTemp</a> | Start Pulling Real-time Temperature Stream |
| <a href="#">IRC_USB_TEMP_CALLBACK</a> | Temperature Callback Function              |
| <a href="#">IRC_USB_StopPullTemp</a>  | Stop Pulling Real-time Temperature Stream  |

### 1) Start Pulling Real-time Temperature Stream

#### IRC\_USB\_StartPullTemp

| Options | Introduction |
|---------|--------------|
|---------|--------------|

|              |                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Description  | Start Pulling Real-time Temperature Stream                                                                                                      |
| Function     | <pre>int IRC_USB_StartPullTemp(     IRC_USB_HANDLE handle,     <a href="#">IRC_USB_TEMP_CALLBACK</a> tempCb,     void* userData);</pre>         |
| Parameter    | <p>param[in] handle Operation handle</p> <p>param[in] tempCb temperature callback function</p> <p>param[in] userData user self-defined data</p> |
| Return Value | <a href="#">status code</a>                                                                                                                     |
| Note         | For specific usage methods, please refer to the demo                                                                                            |

## 2) Temperature Callback Function

### IRC\_USB\_TEMP\_CALLBACK

|             |                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Options     | Introduction                                                                                                                                |
| Description | Temperature callback function                                                                                                               |
| Function    | <pre>int IRC_USB_TEMP_CALLBACK(     IRC_USB_HANDLE handle,     <a href="#">IRC_USB_TEMP_INFO_CB</a>* tempInfo,     void* userData);</pre>   |
| Parameter   | <p>param[in] handle Operation handle</p> <p>param[in] tempCb temperature callback info</p> <p>param[in] userData user self-defined data</p> |

|              |                                                                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Return Value | <a href="#">status code</a>                                                                                                                         |
| Note         | Use in conjunction with“Start Pulling Real-time Temperature Stream”function; The returned temperature frame data format is: Kelvin temperature * 10 |

### 3) Stop Pulling Real-time Temperature Stream

#### IRC\_USB\_StopPullTemp

| Options      | Introduction                                         |
|--------------|------------------------------------------------------|
| Description  | Stop Pulling Real-time Temperature Stream            |
| Function     | int IRC_USB_StopPullTemp(<br>IRC_USB_HANDLE handle); |
| Parameter    | param[in] handle Operation handle                    |
| Return Value | <a href="#">status code</a>                          |
| Note         |                                                      |

### Exception Data Retrieval

| Interface Name                               | Interface Description          |
|----------------------------------------------|--------------------------------|
| <a href="#">IRC_USB_SetExceptionCallback</a> | Global Exception Callback      |
| <a href="#">IRC_USB_EXCEPTION_CALLBACK</a>   | Exception Callback<br>Function |

## 1) Global Exception Callback

### IRC\_USB\_SetExceptionCallback

| Options      | Introduction                                                                                                                                                         |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description  | Global Exception Callback                                                                                                                                            |
| Function     | <pre>int IRC_USB_SetExceptionCallback(<br/>    IRC_USB_HANDLE handle,<br/>    <a href="#">IRC_USB_EXCEPTION_CALLBACK</a> exceptionCb,<br/>    void* userData);</pre> |
| Parameter    | <p>param[in] handle Operation handle</p> <p>param[in] exceptionCb Exception callback function</p> <p>param[in] userData user self-defined data</p>                   |
| Return Value | <a href="#">status code</a>                                                                                                                                          |
| Note         | For specific usage methods, please refer to the demo                                                                                                                 |

## 2) Exception Callback Function

### IRC\_USB\_EXCEPTION\_CALLBACK

| Options     | Introduction                                                                                         |
|-------------|------------------------------------------------------------------------------------------------------|
| Description | Exception callback function                                                                          |
| Function    | <pre>int IRC_USB_EXCEPTION_CALLBACK(<br/>    IRC_USB_HANDLE handle,<br/>    int exceptionType,</pre> |

|              |                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | void* userData);                                                                                                                                                           |
| Parameter    | param[in] handle Operation handle<br>param[in] exceptionCb exception type, refer to<br><a href="#">IRC_USB_EXCEPTION_TYPE</a><br>param[in] userData user self-defined data |
| Return Value | <a href="#">status code</a>                                                                                                                                                |
| Note         | Use in conjunction with“Global Exception<br>Callback”function                                                                                                              |

## Device Configuration

| Interface Name                         | Interface Description                |
|----------------------------------------|--------------------------------------|
| <a href="#">IRC_USB_WritePort</a>      | Writing Serial Port Command          |
| <a href="#">IRC_USB_ReadPort</a>       | Reading Serial Port Command          |
| <a href="#">IRC_USB_CorrectShutter</a> | Shutter correction                   |
| <a href="#">IRC_USB_GetPalleteType</a> | Get Current Color Pallete Index      |
| <a href="#">IRC_USB_SetPalleteType</a> | Set Current Color Pallete Index      |
| <a href="#">IRC_USB_GetTempLevel</a>   | Get Temperature Measurement<br>Range |
| <a href="#">IRC_USB_SeTempLevel</a>    | Set Temperature Measurement<br>Range |
| <a href="#">IRC_USB_GetFPATemp</a>     | Get FPA Temperature                  |

|                                              |                              |
|----------------------------------------------|------------------------------|
| <a href="#"><u>IRC_USB_GetCameraTemp</u></a> | Get Device Temperature       |
| <a href="#"><u>IRC_USB_GetFPASize</u></a>    | Retrive FPA Size             |
| <a href="#"><u>IRC_USB_GetEnvParam</u></a>   | Get Environmental Parameters |
| <a href="#"><u>IRC_USB_SetEnvParam</u></a>   | Set Environmental Parameters |
| <a href="#"><u>IRC_USB_GetTempUnit</u></a>   | Get Temperature Unit         |
| <a href="#"><u>IRC_USB_SetTempUnit</u></a>   | Set Temperature Unit         |
| <a href="#"><u>IRC_USB_GetImageLevel</u></a> | Get Image Enhance Level      |
| <a href="#"><u>IRC_USB_SetImageLevel</u></a> | Set Image Enhance Level      |
| <a href="#"><u>IRC_USB_GetImageParam</u></a> | Get Image Parameters         |
| <a href="#"><u>IRC_USB_SetImageParam</u></a> | Set Image Parameters         |

### 1) Writing Serial Port Command    IRC\_USB\_WritePort

| Options     | Introduction                                                                                                            |
|-------------|-------------------------------------------------------------------------------------------------------------------------|
| Description | Writing Serial Port Command                                                                                             |
| Function    | <pre>int IRC_USB_WritePort(     IRC_USB_HANDLE handle,     char* buf,     int inSize);</pre>                            |
| Parameter   | <p>param[in] handle Operation handle</p> <p>param[in] buf Command Buffer</p> <p>param[in] inSize Write command size</p> |

|              |                                          |
|--------------|------------------------------------------|
| Return Value | <a href="#">status code</a>              |
| Note         | Used to send commands to the core module |

## 2) Reading Serial Port Command IRC\_USB\_ReadPort

| Options      | Introduction                                                                                                                               |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Description  | Reading Serial Port Command                                                                                                                |
| Function     | <pre>int IRC_USB_ReadPort(     IRC_USB_HANDLE handle,     char* buf,     int* outSize,     int timeout);</pre>                             |
| Parameter    | param[in] handle Operation handle<br>param[out] buf Data buffer<br>param[out] outSize Read command size<br>param[out] timeout Read timeout |
| Return Value | <a href="#">status code</a>                                                                                                                |
| Note         | Used to read the commands returned by the core module                                                                                      |

## 3) Shutter correction IRC\_USB\_CorrectShutter

| Options     | Introduction       |
|-------------|--------------------|
| Description | Shutter correction |



|              |                                                        |
|--------------|--------------------------------------------------------|
| Function     | int IRC_USB_CorrectShutter(<br>IRC_USB_HANDLE handle); |
| Parameter    | param[in] handle Operation handle                      |
| Return Value | <a href="#">status code</a>                            |
| Note         |                                                        |

#### 4) Get Current Color Palette Index

##### IRC\_USB\_GetPalleteType

|              |                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------|
| Options      | Introduction                                                                                                                     |
| Description  | Get current color palette index                                                                                                  |
| Function     | int IRC_USB_GetPalleteType(<br>IRC_USB_HANDLE handle,<br>int* palleteType);                                                      |
| Parameter    | param[in] handle Operation handle<br>param[out] palleteType Color Palette type, refer to<br><a href="#">IRC_USB_PALETTE_TYPE</a> |
| Return Value | <a href="#">status code</a>                                                                                                      |
| Note         |                                                                                                                                  |

#### 5) Set Color Palette Index IRC\_USB\_SetPalleteType

|             |                         |
|-------------|-------------------------|
| Options     | Introduction            |
| Description | Set color palette index |

|              |                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| Function     | int IRC_USB_SetPalleteType(<br>IRC_USB_HANDLE handle,<br>int* palleteType);                                                     |
| Parameter    | param[in] handle Operation handle<br>param[in] palleteType Color Palette type, refer to<br><a href="#">IRC_USB_PALETTE_TYPE</a> |
| Return Value | <a href="#">status code</a>                                                                                                     |
| Note         |                                                                                                                                 |

## 6) Get Temperature Measurement Range

### IRC\_USB\_GetTempLevel

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Options      | Introduction                                                                                                                                     |
| Description  | Get temperature measurement range                                                                                                                |
| Function     | int IRC_USB_GetTempLevel(<br>IRC_USB_HANDLE handle,<br>int* gainType);                                                                           |
| Parameter    | param[in] handle Operation handle<br>param[in] gainType level temperature measurement<br>range, refer to <a href="#">IRC_USB_TEMP_LEVEL_TYPE</a> |
| Return Value | <a href="#">status code</a>                                                                                                                      |
| Note         |                                                                                                                                                  |

## 7) Set Temperature Measurement Range

### IRC\_USB\_SeTempLevel

| Options      | Introduction                                                                                                                                     |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Description  | Set temperature measurement range                                                                                                                |
| Function     | int IRC_USB_SeTempLevel(<br>IRC_USB_HANDLE handle,<br>int gainType);                                                                             |
| Parameter    | param[in] handle Operation handle<br>param[in] gainType level temperature measurement<br>range, refer to <a href="#">IRC_USB_TEMP_LEVEL_TYPE</a> |
| Return Value | <a href="#">status code</a>                                                                                                                      |
| Note         |                                                                                                                                                  |

## 8) Get FPA Temperature IRC\_USB\_GetFPATemp

| Options     | Introduction                                                                          |
|-------------|---------------------------------------------------------------------------------------|
| Description | Get FPA Temperature                                                                   |
| Function    | int IRC_USB_GetFPATemp(<br>IRC_USB_HANDLE handle,<br>float* fpaTemp);                 |
| Parameter   | param[in] handle Operation handle<br>param[out] fpaTemp FPA temperature, celsius * 10 |

|              |                             |
|--------------|-----------------------------|
| Return Value | <a href="#">status code</a> |
| Note         |                             |

## 9) Get Device Temperature IRC\_USB\_GetCameraTemp

| Options      | Introduction                                                                                |
|--------------|---------------------------------------------------------------------------------------------|
| Description  | Get device temperature                                                                      |
| Function     | <pre>int IRC_USB_GetCameraTemp(     IRC_USB_HANDLE handle,     float* fpaTemp);</pre>       |
| Parameter    | param[in] handle Operation handle<br>param[out] cameraTemp Device temperature, celsius * 10 |
| Return Value | <a href="#">status code</a>                                                                 |
| Note         |                                                                                             |

## 10) Retrieve FPA Size IRC\_USB\_GetFPASize

| Options     | Introduction                                                                                    |
|-------------|-------------------------------------------------------------------------------------------------|
| Description | Retrive FPA size                                                                                |
| Function    | <pre>int IRC_USB_GetFPASize(     IRC_USB_HANDLE handle,     int* width,     int* height);</pre> |

|              |                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------|
| Parameter    | param[in] handle Operation handle<br>param[out] width FPA width<br>param[out] height FPA height |
| Return Value | <a href="#">status code</a>                                                                     |
| Note         |                                                                                                 |

## 11)Get Environment Parameters    IRC\_USB\_GetEnvParam

| Options      | Introduction                                                                                                     |
|--------------|------------------------------------------------------------------------------------------------------------------|
| Description  | Get environmental parameters                                                                                     |
| Function     | <pre>int IRC_USB_GetEnvParam(     IRC_USB_HANDLE handle,     <a href="#">IRC_USB_ENV_PARAM</a>* envParam);</pre> |
| Parameter    | param[in] handle Operation handle<br>param[out] envParam environmental parameters                                |
| Return Value | <a href="#">status code</a>                                                                                      |
| Note         |                                                                                                                  |

## 12)    Set Environment Parameters    IRC\_USB\_SetEnvParam

| Options     | Introduction                                                   |
|-------------|----------------------------------------------------------------|
| Description | Set Environmental Parameters                                   |
| Function    | <pre>int IRC_USB_SetEnvParam(     IRC_USB_HANDLE handle,</pre> |

|              |                                                                                  |
|--------------|----------------------------------------------------------------------------------|
|              | const <a href="#">IRC_USB_ENV_PARAM</a> * envParam);                             |
| Parameter    | param[in] handle Operation handle<br>param[in] envParam environmental parameters |
| Return Value | <a href="#">status code</a>                                                      |
| Note         |                                                                                  |

### 13) Get Temperature Unit IRC\_USB\_GetTempUnit

|              |                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Options      | Introduction                                                                                                             |
| Description  | Get temperature unit                                                                                                     |
| Function     | int IRC_USB_GetTempUnit(<br>IRC_USB_HANDLE handle,<br>int* tempUnit);                                                    |
| Parameter    | param[in] handle Operation handle<br>param[out] tempUnit temperature unit, refer to<br><a href="#">IRC_USB_TEMP_UNIT</a> |
| Return Value | <a href="#">status code</a>                                                                                              |
| Note         |                                                                                                                          |

### 14) Set Temperature Unit IRC\_USB\_SetTempUnit

|             |                              |
|-------------|------------------------------|
| Options     | Introduction                 |
| Description | Set temperature unit         |
| Function    | int IRC_USB_SetTempUnit(<br> |

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
|              | IRC_USB_HANDLE handle,<br>int tempUnit);                                                                                |
| Parameter    | param[in] handle Operation handle<br>param[in] tempUnit temperature unit, refer to<br><a href="#">IRC_NET_TEMP_UNIT</a> |
| Return Value | <a href="#">status code</a>                                                                                             |
| Note         |                                                                                                                         |

### 15) Get Image Enhance Level IRC\_USB\_GetImageLevel

| Options      | Introduction                                                                                         |
|--------------|------------------------------------------------------------------------------------------------------|
| Description  | Get image enhance level                                                                              |
| Function     | int IRC_USB_GetImageLevel(<br>IRC_USB_HANDLE handle,<br>int* level);                                 |
| Parameter    | param[in] handle Operation handle<br>param[out] level Enhance Level; 0: manual, 0-9: preset<br>level |
| Return Value | <a href="#">status code</a>                                                                          |
| Note         |                                                                                                      |

### 16) Set Image Enhance Level IRC\_USB\_SetImageLevel

| Options | Introduction |
|---------|--------------|
|---------|--------------|

|              |                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------|
| Description  | Set image enhance level                                                                          |
| Function     | int IRC_USB_SetImageLevel(<br>IRC_USB_HANDLE handle,<br>int level);                              |
| Parameter    | param[in] handle Operation handle<br>param[in] level Enhance Level; 0: manual, 0-9: preset level |
| Return Value | <a href="#">status code</a>                                                                      |
| Note         |                                                                                                  |

## 17) Get Image Parameters IRC\_USB\_GetImageParam

|              |                                                                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Options      | Introduction                                                                                                                                           |
| Description  | Get image parameters                                                                                                                                   |
| Function     | int IRC_USB_GetImageParam(<br>IRC_USB_HANDLE handle,<br>int paramType,<br>int* value);                                                                 |
| Parameter    | param[in] handle Operation handle<br>param[out] paramType image parameters type, refer to <a href="#">IRC_USB_IMAGE_PARAM_TYPE</a><br>param[out] value |
| Return Value | <a href="#">status code</a>                                                                                                                            |



|      |  |
|------|--|
| Note |  |
|------|--|

## 18) Set Image Parameter IRC\_USB\_SetImageParam

| Options      | Introduction                                                                                                                                                        |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description  | Set Image Parameter                                                                                                                                                 |
| Function     | <pre>int IRC_USB_SetImageParam(     IRC_USB_HANDLE handle,     int paramType,     int value);</pre>                                                                 |
| Parameter    | <p>param[in] handle Operation handle</p> <p>param[in] paramType image parameters type, refer to <a href="#">IRC_USB_IMAGE_PARAM_TYPE</a></p> <p>param[in] value</p> |
| Return Value | <a href="#">status code</a>                                                                                                                                         |
| Note         |                                                                                                                                                                     |

# Chapter 4 Structure Introduction

## 1. Device Information IRC\_USB\_DEV\_INFO

### Structure Definition

```
typedef struct
{
    int id; ///< Device ID

    char devName[IRC_USB_NAME_LEN_MAX]; ///< Device name

    char portName[IRC_USB_NAME_LEN_MAX]; ///< Serial port
name
}
```

## 2. Environment Parameter IRC\_USB\_ENV\_PARAM

### Structure Definition

```
typedef struct
{
    float atmosphereTemp; ///< atmospheric temperature

    float distance; ///< target distance

    float emissivity; ///< emissivity 0.01-1

    float reflectedTemp; ///< reflected temperature

    float transmittance; ///< atmospheric transmissivity 0.01-1
```

```
}
```

### 3. Video Callback Information

#### IRC\_USB\_VIDEO\_INFO\_CB

##### Structure Definition

```
typedef struct
{
    char* dataBuf; ///< video frame data

    int width; ///< width

    int height; ///< height

    int frameFmt; ///< video frame format, refer to
IRC\_USB\_FRAME\_FMT
}
```

### 4. Temperature Callback Information

#### IRC\_USB\_TEMP\_INFO\_CB

##### Structure Definition

```
typedef struct
{
    char* dataBuf; ///< temperature frame data

    int width; ///< width

    int height; ///< height
}
```

}

# Chapter 5 Constant

## 1. status code list IRC\_USB\_ERROR

| Definition                                                                                                                                                                                                                           | Description |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <pre>typedef enum {     IRC_USB_ERROR_OK = 0, ///&lt; succeed     IRC_USB_ERROR_FAILED = 1, ///&lt; fail     IRC_USB_ERROR_NOT_SUPPORTED = 2, ///&lt; unavailable     IRC_USB_ERROR_PARAM_WRONG = 3, ///&lt; parameter error }</pre> |             |

## 2. Type of Global Exception

### IRC\_USB\_EXCEPTION\_TYPE

| Definition                                                                                                                                                         | Description |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <pre>typedef enum {     IRC_USB_EXCEPTION_DEV_OFFLINE = 1001, ///&lt; device offline     IRC_USB_EXCEPTION_PREVIEW_OFFLINE = 1002, ///&lt; preview offline }</pre> |             |

```

        IRC_USB_EXCEPTION_TEMP_OFFLINE = 1003, ///  

temperature offline  

    }

```

### 3. Level temperature measurement range

#### IRC\_USB\_TEMP\_LEVEL\_TYPE

| Definition Description |
|------------------------|
|------------------------|

|                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> typedef enum<br/> {<br/>     IRC_USB_TEMP_LEVEL_HG = 0, ///<br/>     IRC_USB_TEMP_LEVEL_LG, ///<br/>     IRC_USB_TEMP_LEVEL_AUTO ///<br/> } </pre> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|

### 4. Color Palette type IRC\_USB\_PALETTE\_TYPE

| Definition Description |
|------------------------|
|------------------------|

|                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> typedef enum<br/> {<br/>     IRC_USB_PALETTE_WHITE_HOT = 0, ///<br/>     IRC_USB_PALETTE_BLACK_HOT, ///<br/>     IRC_USB_PALETTE_RAINBOW, ///<br/>     IRC_USB_PALETTE_RAINBOW_HC, ///<br/> } </pre> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

```

IRC_USB_PALETTE_IRON, ///< Iron
IRC_USB_PALETTE_LAVA, ///< Lava
IRC_USB_PALETTE_SKY, ///< Sky
IRC_USB_PALETTE_MID_GREY, ///< MidGrey
IRC_USB_PALETTE_RDGY, ///< RdGy
IRC_USB_PALETTE_PUOR, ///< PuOr
IRC_USB_PALETTE_SPECIAL, ///< Special
IRC_USB_PALETTE_RED, ///< Red
IRC_USB_PALETTE_ICE_FIRE, ///< IceFire
IRC_USB_PALETTE_GREE_RED, ///< GreenRed
IRC_USB_PALETTE_SPECIAL2, ///< Special2
IRC_USB_PALETTE_RED_HOT, ///< RedHot
IRC_USB_PALETTE_GREEN_HOT, ///< GreenHot
IRC_USB_PALETTE_BLUE_HOT, ///< BlueHot
IRC_USB_PALETTE_GREEN, ///< Green
IRC_USB_PALETTE_BLUE, ///< Blue
}

```

## 5. Temperature unit IRC\_USB\_TEMP\_UNIT

| Definition   | Description |
|--------------|-------------|
| typedef enum |             |
| {            |             |

```

    IRC_USB_TEMP_CENTIGRADE = 0, ///< Celsius

    IRC_USB_TEMP_KELVIN, ///< Kelvin

    IRC_USB_TEMP_FAHRENHEIT ///< Fahrenheit
}

```

## 6. Frame Forma IRC\_USB\_FRAME\_FMT

| Definition                                            | Description |
|-------------------------------------------------------|-------------|
| typedef enum                                          |             |
| {                                                     |             |
| IRC_NET_FRAME_FMT_YUYV = 0, ///< Imaging device image |             |
| format                                                |             |
| IRC_NET_FRAME_FMT_UYVY, ///< Temping device image     |             |
| format                                                |             |
| }                                                     |             |

## 6. Image Parameter Type

### IRC\_USB\_IMAGE\_PARAM\_TYPE

| Definition                                    | Description |
|-----------------------------------------------|-------------|
| typedef enum                                  |             |
| {                                             |             |
| IRC_USB_IMAGE_PARAM_TYPE_BRIGHTNESS = 0, ///< |             |
| Brightness                                    |             |



IRC\_USB\_IMAGE\_PARAM\_TYPE\_CONTRAST, ///  
Contrast

IRC\_USB\_IMAGE\_PARAM\_TYPE\_DETAIL, ///  
Detail

Enhancement

IRC\_USB\_IMAGE\_PARAM\_TYPE\_SPATIAL\_DENOISING, ///  
Spatial Domain Denoising

Spatial Domain Denoising

IRC\_USB\_IMAGE\_PARAM\_TYPE\_TIME\_DENOISING, ///  
Temporal Domain Denoising

Temporal Domain Denoising

IRC\_USB\_IMAGE\_PARAM\_TYPE\_DYNAMIC\_RANGE, ///  
Dynamic Range

Dynamic Range

}

# Chapter 6 Demo Source Code & Operation Guide

## 1. Demo Source Code

### 1) Compiling Environment

Windows: win10, VS2019 + Qt5.15.2

Linux: ubuntu18.04, Qt5.15.2

### 2) QT Configuration Instructions

Reference: [Windows系统安装Qt 5.15.2完全版 Qt5.15.2-CSDN博客](#)

### 3) Windows C++ Basic Runtime Library Description

The Microsoft runtime libraries are used during the Windows SDK process.

Download path:

### 4) Folder Content Description

|               | Source Files Location | Library Files Location   | Header Files Location |
|---------------|-----------------------|--------------------------|-----------------------|
| Windows/Linux | ../Demo/C++/src       | ../SDK/x64    ../SDK/x86 | ../SDK/include        |

### 5) SDK Update Description

To update the SDK, you only need to update the infrared libraries and header files.

Windows:

Infrared libraries are IRCUSBSdk.dll, IRCUSBSdk.lib; other libraries are the dependencies used by the infrared library.

Header files are IRCUSBSdk.h, IRCUSBSdkDef.h.

Linux:

Infrared libraries are LibIRCUSBSdk.so, LibIRCUSBSdk.so.1, LibIRCUSBSdk.so.1.0, LibIRCUSBSdk.so.1.0.0;

Header files are IRCUSBSdk.h, IRCUSBSdkDef.h.

## 2. Demo Operation

The demo is divided into five modules for demonstration: 1. Device Connection and Streaming, 2. Image, 3. Temp, 4. Analysis (to be supplemented later) and 5. General.

### 1. Device Connection and Streaming

Corresponds to the "Device Connection", "Live Preview" and "Temperature Data Retrieval" sections in the manual, demonstrating interfaces such as "Open device" and "Start pulling Real-time Temperature Stream".

### 2. Image

Corresponds to the "Device Configuration" section in the manual, involving image-related settings such as "Set Current Color Palette Index".

### 3. Temp

Corresponds to the "Device Configuration" section in the manual, involving temperature measurement settings such as "Set Environment Parameters".

### 4. Analysis

To be supplemented later.

### 5. General

Corresponds to the "Device Configuration" section in the manual, involving basic configuration settings such as "Retrieve FPA Size".